



COURSE SYLLABUS						
Cours e Code	COMP 006	Course Title	Data Structure and Algorithms		Course Credit	3
Course Description		This course provides a comprehensive study of data organization and algorithm design essential for problem-solving in computer science and information technology. It focuses on analyzing and applying efficient data structures such as arrays, linked lists, stacks, queues, trees, hash tables, heaps, tries, and graphs in developing real-world solutions. Students will learn to evaluate algorithm performance, optimize programs, and apply these techniques in practical, collaborative, and ethical software development contexts.				
Pre-Requisites			Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



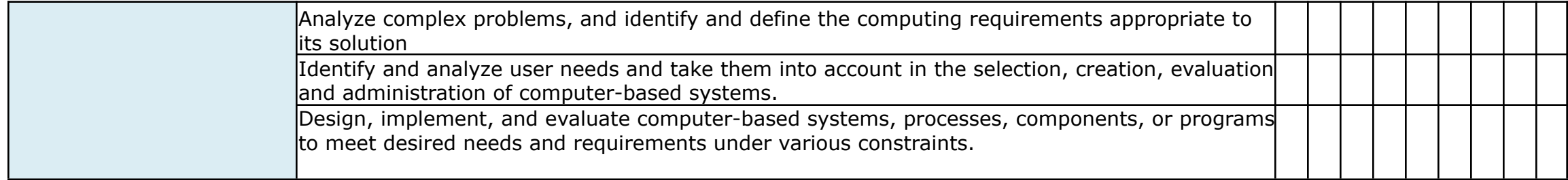
**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9



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	Integrate IT-based solutions into the user environment effectively.												
	Assist in the creation of an effective IT project plan.												
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession												
	Understand best practices and standards and their applications.												
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.												
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.												
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.												
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.												
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development												

COURSE LEARNING OUTCOMES (CLO)	Explain the role and importance of data structures and algorithms in computing and real-world applications.	Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Identify and differentiate various data structures and their appropriate use cases.								

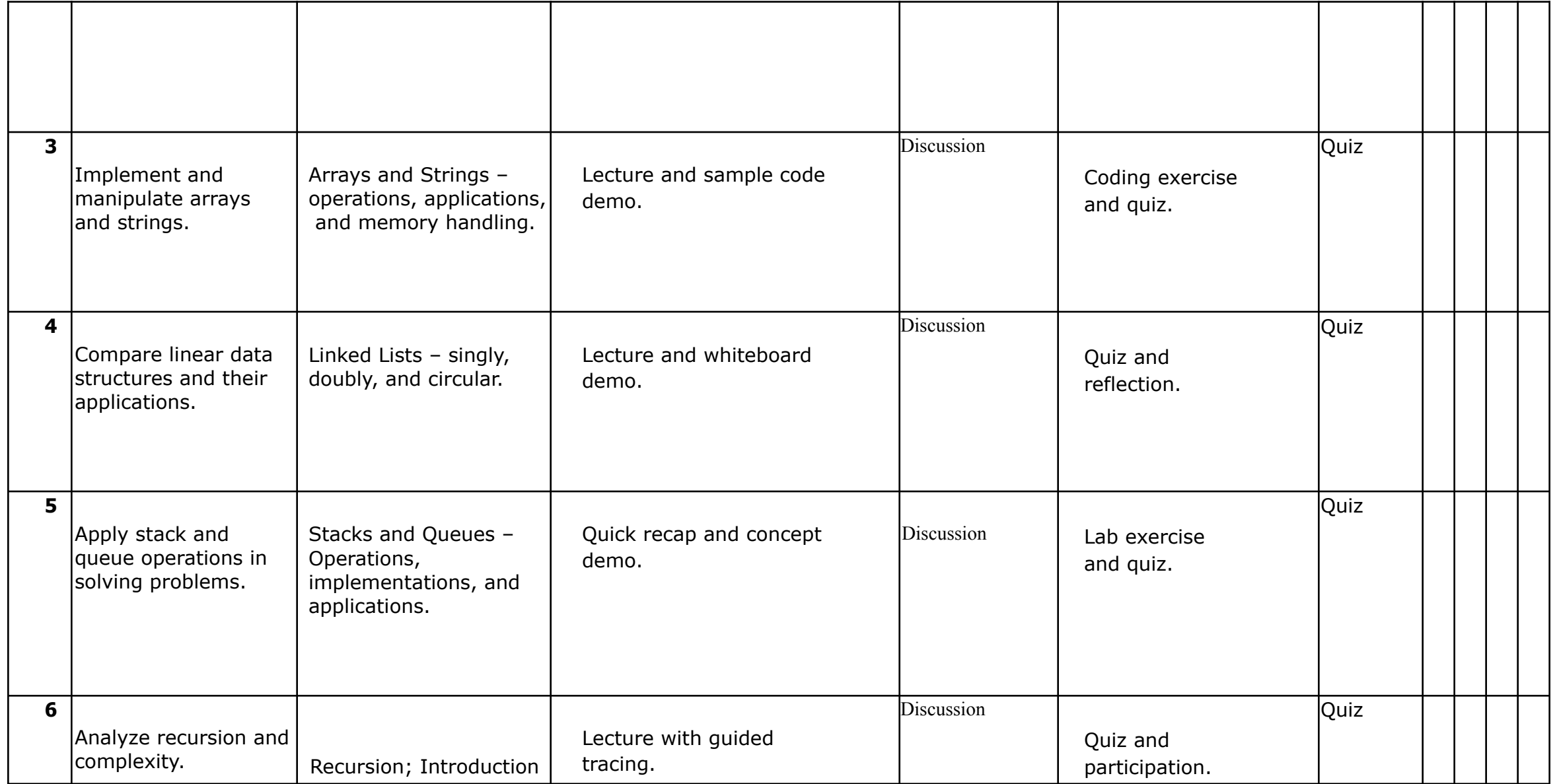


	Implement data structures and algorithms using a high-level programming language.								
	Analyze algorithm efficiency in terms of time and space complexity.								
	Design and develop solutions using appropriate data structures and algorithmic strategies.								
	Apply advanced data structures such as hash tables, heaps, and tries to optimize performance.								
	Demonstrate collaborative, ethical, and professional practices in algorithm design and implementation.								
	Evaluate and adapt algorithmic solutions for scalability, reliability, and performance in industry contexts.								



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
							Face-to-Face	FLTAs		
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	Explain the importance of algorithms and structured problem-solving.	Introduction to Algorithms; Problem-solving steps; Flowcharts and Pseudocode.	Lecture with visual examples.	Discussion	Quiz and participation.	Quiz				

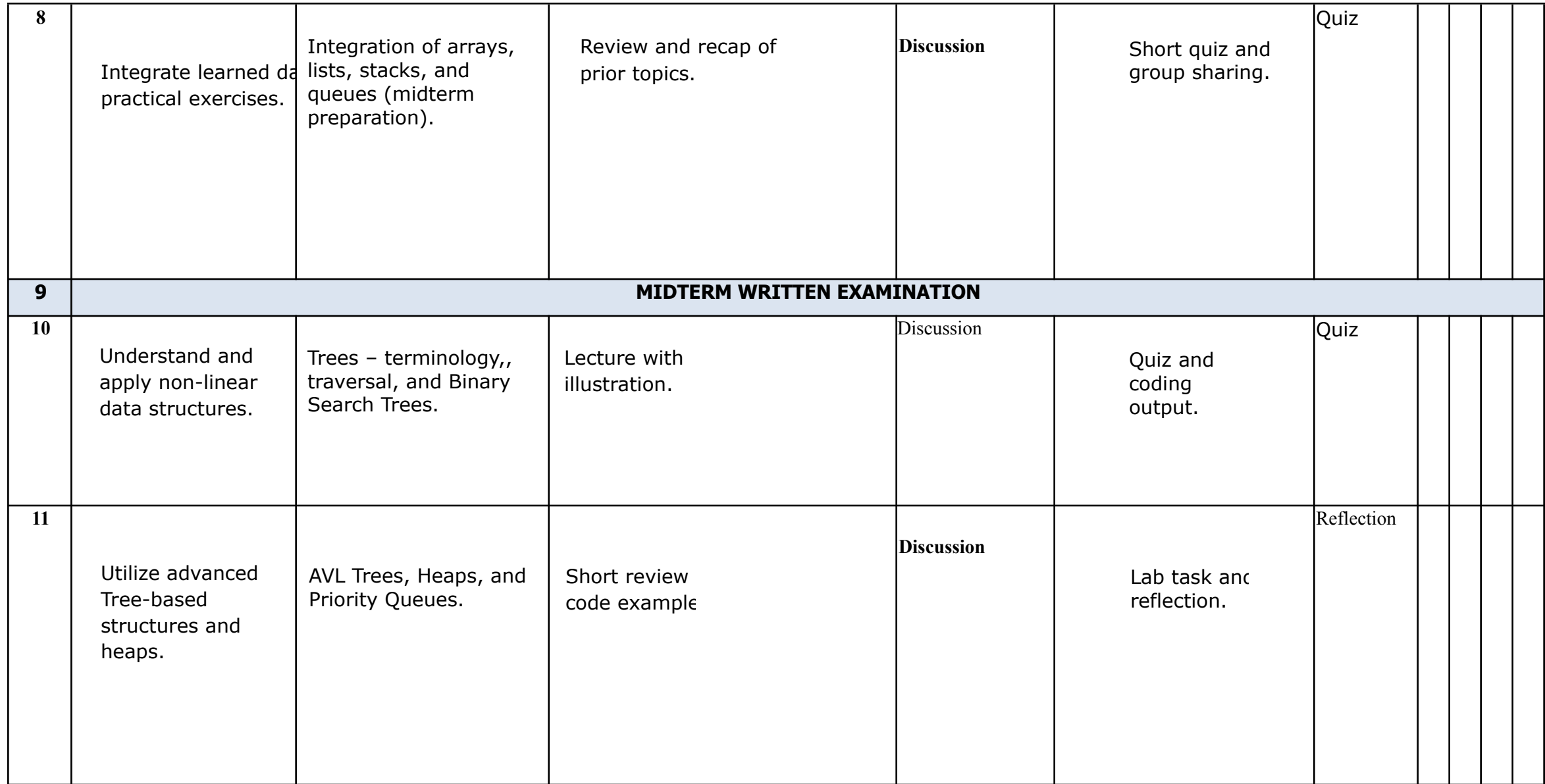


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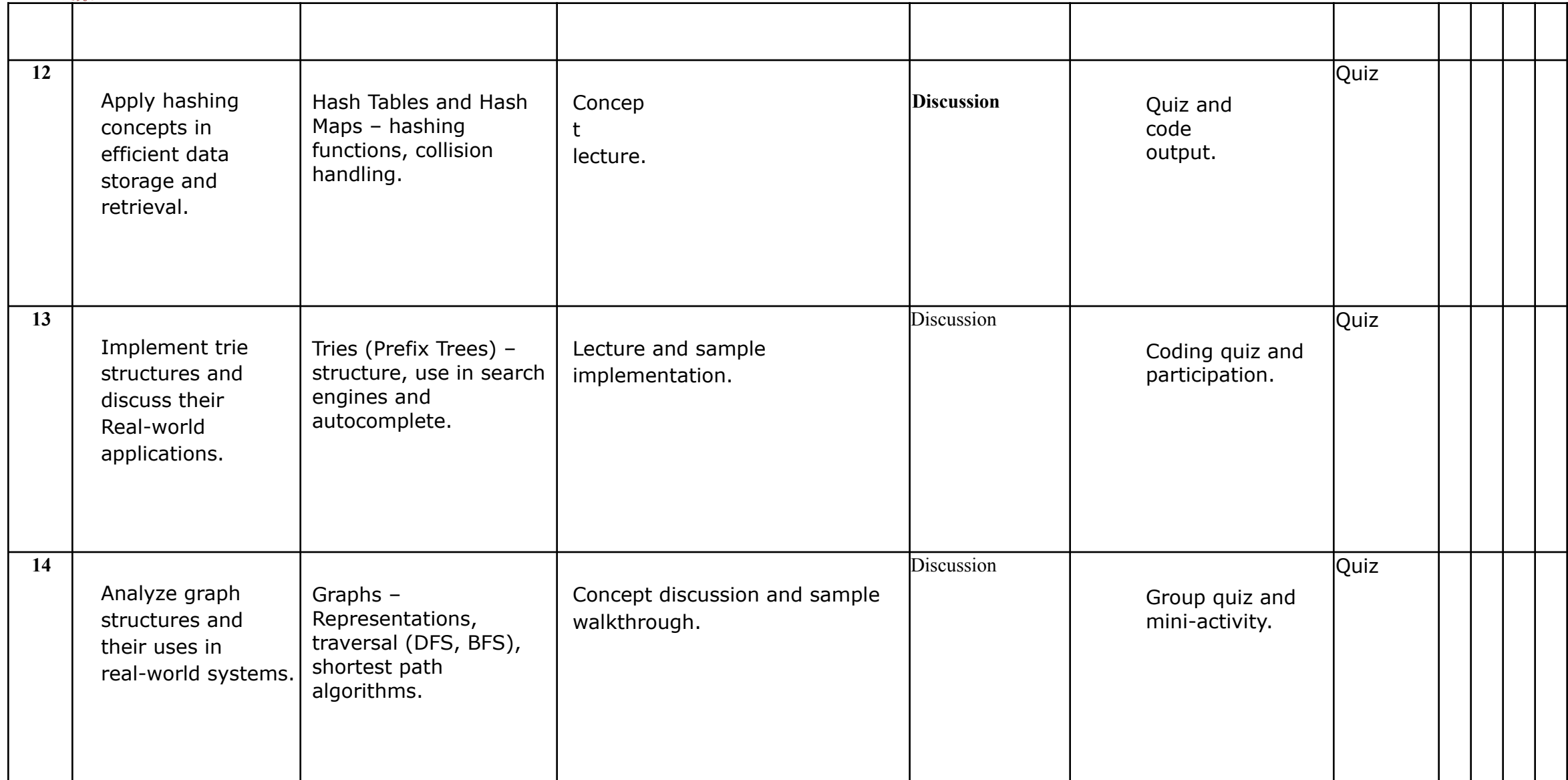


		to Big O Notation – time and space complexity.								
7	Implement and evaluate algorithms.	Searching (Linear, Binary) and Sorting (Bubble, Insertion, Merge, Quick).	Discussion and board work.	Discussion	Coding quiz and activity.	Quiz				



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15	Apply dynamic programming optimization problems.	Dynamic Programming (DP) strategies.	Lecture and problem-solving discussion.	Discussion	Quiz and short coding exercise.	Quiz				
16	Design and develop multiple data structures.	Project planning, development DSA-based solution.	Instructor coaching and feedback.	Discussion	Project proposal and presentation draft.	Project proposal and presentation draft.				
17	Demonstrate collaborative algorithm development.	Professional ethics, team software projects.	Guided discussion and peer sharing.	Discussion	Peer evaluation and reflection paper.	Peer evaluation				



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18	FINAL WRITTEN EXAMINATION										



		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies ● Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		● Quizzes					
18	FINAL EXAMINATION									
LIST OF REFERENCES BOOKS: WEBSITE: GeeksforGeeks. (2025, October 4). <i>DSA Tutorial – Learn data structures and algorithms</i> . https://www.geeksforgeeks.org/data-structures/ Programiz. (n.d.). <i>Learn data structures and algorithms</i> . https://www.programiz.com/dsa W3Schools. (n.d.). <i>Data structures and algorithms tutorial</i> . https://www.w3schools.com/dsa/index.php VisuAlgo. (n.d.). <i>Visualising data structures & algorithms (site)</i> . https://visualgo.net/en_visual										



2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



The WORLD
UNIVERSITY
RANKINGS
for INNOVATION

Rank 21
GENERATIVE AI
Rank 39
STUDENT SUPPORT
Rank 50
SYMBOL/



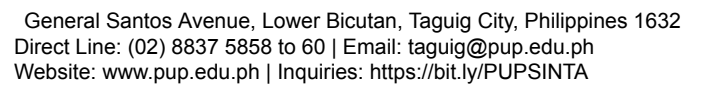
5 stars
Employability
Teaching
Academic
Development
Inclusiveness
Social Responsibility
Specialist Criteria: Bachelor
of Science in Nutrition and
Dietetics (BSND)



Times Higher Education
Impact Rankings 2024
DEBUTS AT 1501+



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



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The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will



	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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Mid Term		Final Term	
Class Standing	70%	Class Standing	70%
Exercises/Activities		Exercises/Activities	
Summative Quizzes		Summative Quizzes	
Recitation and Attendance		Recitation and Attendance	
Reflection Paper		Reflection Paper	
Midterm Examinations	<u>30%</u>	Final Examinations	30%
	100%		100%

	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.	
COURSE REQUIREMENT/S	GRADING SYSTEM	
- Completion of quizzes, laboratory exercises, and a final project (focusing on accuracy, logical implementation, and application of data structures and algorithms).	Final Grade = (Midterm Grade + Final Term Grade) / 2	



CLASS INFORMATION		FACULTY INFORMATION	
Course, Year and Section: BSIT, 2-1		Name of Faculty: Francis F. Franco	
Day and Time:		Consultation Time:	
Room:		Office Tel. No./ Mobile Phone No.	
School Year:	Semester:	E-Mail Address:	

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
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